# AI Adoption in Higher Education: A Meta-Analytic Structural Equation Modeling Study

Target journal: Computers & Education

Draft date: 2026-06-12

## Submission Package State

This target-journal draft updates the existing APA-style shell for Computers & Education. It includes the introduction, methods, input/QC results, table/figure spine, and team insertion points. It deliberately does not invent Stage 1/Stage 2 estimates. The lead-approved TSSEM/OSMASEM run remains gated by Paper A numeric sample-size reconciliation.

## Highlights

- Synthesizes AI adoption evidence in higher education with MASEM.
- Integrates TAM/UTAUT predictors with trust and AI anxiety.
- Separates direct-r inputs from converted sensitivity evidence.
- Tests structural and moderator paths across ten constructs.
- Provides reproducible extraction and QC artifacts.

## Abstract

Artificial intelligence tools are increasingly embedded in higher education, but the empirical adoption literature remains fragmented across constructs, samples, tools, and reporting formats. This study synthesizes higher-education AI adoption evidence using meta-analytic structural equation modeling. The planned model integrates performance expectancy, effort expectancy, social influence, facilitating conditions, attitude, self-efficacy, trust in AI, AI anxiety, behavioral intention, and use behavior. The current analysis-ready package preserves source-reported direct correlations separately from expanded direct-r-form and converted sensitivity evidence. At the 2026-06-05 checkpoint, the model-ready primary input contains 804 rows after tiered source decisions. The current target-journal package documents the final structure, tables, figures, and analysis gate for a Computers & Education submission. Final path estimates, indirect effects, model fit, and moderator results should be inserted only after sample-size reconciliation and the approved TSSEM/OSMASEM run are complete.

Keywords: artificial intelligence; technology acceptance; higher education; MASEM; UTAUT; trust; anxiety

## Introduction

Artificial intelligence tools are now embedded in higher education through large language models, intelligent tutoring systems, automated assessment systems, writing assistants, recommendation tools, and analytics platforms. Their spread has produced a rapidly expanding empirical literature on adoption, acceptance, and use. Yet this literature remains difficult to interpret cumulatively because studies draw from overlapping but nonidentical acceptance frameworks, measure different subsets of constructs, and report evidence in formats that do not directly support a single structural synthesis.

Meta-analytic structural equation modeling is well suited to this problem because it can synthesize study-level correlation matrices and test a theory-guided network of relationships. In AI adoption research, this is especially important. Performance expectancy, effort expectancy, social influence, facilitating conditions, attitude, self-efficacy, behavioral intention, and use behavior provide continuity with TAM, TPB, and UTAUT. Trust in AI and AI anxiety capture psychological features of AI systems that are not fully reducible to general usefulness or ease-of-use beliefs.

The present study develops a MASEM of AI adoption in higher education that integrates traditional technology acceptance constructs with AI-specific psychological constructs. The working model treats attitude as a theoretically meaningful mediator rather than assuming that the parsimonious UTAUT exclusion of attitude applies unchanged to AI adoption contexts. It also tests whether trust and anxiety contribute to behavioral intention beyond standard acceptance predictors.

## Literature Review

[Reserved for team contribution. Use the Team Writing Brief in docs/07_manuscript_exemplars/20260612/TEAM_WRITING_BRIEF_LIT_REVIEW_DISCUSSION_20260612.md.]

## Method

### Design and Reporting

Paper A is the parent meta-analysis for the AI adoption evidence-synthesis project. It uses systematic-review procedures to identify eligible studies and applies TSSEM/OSMASEM to synthesize construct-level relationships. Reporting should align with PRISMA 2020 and Computers & Education submission requirements.

### Search, Screening, and Eligibility

The documented search workflow yielded 22,166 records. After deduplication, 16,189 records remained for screening. The final manuscript must harmonize the proposal-stage included-study count with the final locked full-text MASEM-eligible count before submission.

### Constructs and Model Architecture

The primary model uses 10 constructs: performance expectancy, effort expectancy, social influence, facilitating conditions, attitude, self-efficacy, trust in AI, AI anxiety, behavioral intention, and use behavior. The planned architecture places performance expectancy and effort expectancy upstream of attitude, attitude upstream of behavioral intention, behavioral intention upstream of use behavior, and trust/anxiety as AI-specific antecedents of behavioral intention.

### Analysis Plan

The primary analysis will use two-stage MASEM after numeric sample-size reconciliation. Stage 1 will pool study-level correlation matrices with random-effects TSSEM when feasible. Stage 2 will fit the prespecified structural model to the pooled matrix. OSMASEM or equivalent meta-regression will test moderators when moderator data are sufficiently complete. Sensitivity analyses will separate primary direct-r evidence from expanded direct-r-form and converted beta/path/source-statistic evidence.

## Results

### Analysis-Ready Input

| Input or gate | Current value | Submission interpretation |
|---|---|---|
| Primary model-ready rows | 804 | Main candidate input after tiered freeze |
| Usable rows after r checks | 796 | Rows available for matrix construction once N is resolved |
| Rows missing numeric N | 754 | Primary N-weighted SEM blocker |
| Studies represented | 74 | Current study-level matrix universe |
| Construct-pair coverage | 44/45 | Broad but incomplete coverage |

### Primary MASEM Results

[Lead insertion point. Insert Stage 1 pooled correlation matrix, heterogeneity estimates, Stage 2 path coefficients, indirect effects, model fit, and sensitivity results after the Paper A sample-size gate and TSSEM/OSMASEM run are complete.]

## Discussion

[Reserved for team contribution after lead inserts final results.]